THE COTANGENT COMPLEX OF A MORPHISM TO A CONSTANT ALGEBRAIC SPACE AS A LOWER SHRIEK OF DIFFERENTIALS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays a chain of three equalities; the proof justifies the second and third first and then identifies the first with the cotangent complex of the ring map.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The computation of the lower shriek as sheafification of an explicit presheaf complex. Remark remark-compute-L-pi-shriek-spaces, used at proof lines 4727-4740. Describes Lpi_! of the sheaf of differentials.
- The section-level map identifying the complex with the ring-level cotangent complex. Remark remark-map-sections-over-U, used at proof lines 4740-4744. Identifies the complex as computing the cotangent complex of Lambda -> O_X.
- The description of the cotangent complex of an algebraic space over a ring. Lemma lemma-space-over-ring, used at proof lines 4744-4745. Concludes the identification with L_{X/Lambda}.
- Flatness of the module of differentials for the objects of the category. used at proof lines 4718-4724. Gives the second and third equalities of the displayed chain.

Source reference. Original lemma 08VG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11234–11234. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 08VL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11239–11239. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 08SX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11145–11145. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *In the situation above there is a canonical isomorphism*

$$L_{X/\Lambda} = L\pi_!(Li^*\Omega_{\mathcal{O}/\underline{\Lambda}}) = L\pi_!(i^*\Omega_{\mathcal{O}/\underline{\Lambda}}) = L\pi_!(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X)$$

in $D(\mathcal{O}_X)$.

Proof. We first observe that for any object $(U \to \mathbf{A})$ of $\mathcal{C}_{X/\Lambda}$ the value of the sheaf $\mathcal{O}$ is a polynomial algebra over Λ. Hence $\Omega_{\mathcal{O}/\underline{\Lambda}}$ is a flat $\mathcal{O}$-module and we conclude the second and third equalities of the statement of the lemma hold.

By Remark 08VL the object $L\pi_!(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X)$ is computed as the sheafification of the complex of presheaves

$$U \mapsto \left(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X\right)(\mathbf{A}_{\bullet,U}) = \Omega_{P_{\bullet,U}/\Lambda} \otimes_{P_{\bullet,U}} \mathcal{O}_X(U) = L_{\mathcal{O}_X(U)/\Lambda}$$

using notation as in Remark 08VL. Now Remark 08SX shows that $L\pi_!(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X)$ computes the cotangent complex of the map of rings $\underline{\Lambda} \to \mathcal{O}_X$ on $X_{étale}$. This is what we want by Lemma 08VG. □